within the MTS method by a modified potential function $\bar{U}(\boldsymbol{q}) \stackrel{\text{def}}{=} U_{\text{S}}(\mathcal{A}(\boldsymbol{q}))$. The mapping $\mathcal{A}(\boldsymbol{q})$ is defined by reference to the fast dynamics, e.g. by the weighted average

$$\mathcal{A}(\boldsymbol{q}) = \frac{1}{K+1} \sum_{i=0}^{K} \phi_i \boldsymbol{\theta}_i(\boldsymbol{q}),$$

where the $\boldsymbol{\theta}_i$ are defined recursively by solving the system with Hamiltonian

$$H^{\text{fast}}(\boldsymbol{q}, \boldsymbol{p}) = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U_{\text{F}}(\boldsymbol{q})$$

starting from initial position $\boldsymbol{q}$ and zero initial momentum $\boldsymbol{p}$, for K timesteps using a small stepsize τ_f. This small step does not have to be directly related to the timestep used for integrating the fast dynamics within the MTS scheme; it is only used for the averaging procedure. The resulting sequence of $\boldsymbol{\theta}_i$ clearly are functions of $\boldsymbol{q}$ only and so $\mathcal{A}$ is well defined. The weights ϕ_i appearing in the average may be selected in various ways. The initialization of the calculation of the $\boldsymbol{\theta}_i$ can, moreover, be modified by various alternative $\boldsymbol{q}$-dependent proposals, or may be randomized.

An alternative, simpler, and probably more robust, procedure is simply to fix the value of $\mathcal{A}(\boldsymbol{q})$ by constraining $\boldsymbol{q}$ to its mechanical equilibrium value [189] in which case no fast averaging whatsoever needs to be performed. This approach relies on the use of the SHAKE method of constrained molecular dynamics (see the following section).

In each of the mollified multiple timestepping methods, since only the potential energy is modified, and by a smooth mapping of positions, the forces derived by differentiating $\bar{U}$ are still conservative, and, just as with the MTS method, the scheme remains symplectic.

The mollification is only applied for the purpose of modifying the slow impulse and the mollified methods are in other ways identical to multiple timestepping. These schemes appear to confer some distinct advantage in the large stepsize in molecular dynamics. For example where a detailed and fully flexible molecular model of a solvated biomolecule might require stepsizes of 2 fs in a Verlet simulation, mollified impulse methods may allow increases to 6 fs [189] in the 'large' timestep, i.e. the period between reevaluations of the slow forces. This benefit may be substantial in simulations of large systems.

4.3 Methods for Constrained Dynamics

Constraints are used frequently in molecular modelling. By a constraint, we mean a modification of a dynamical system to maintain the constancy of a given function of coordinates (or coordinates and momenta). In this section we consider *holonomic* constraints which may be expressed in terms of positions only, and perhaps time,

i.e.

$$g(\boldsymbol{q}, t) = 0. \tag{4.3}$$

Note that by differentiating (4.3) with respect to time, we obtain $\nabla_{\boldsymbol{q}} g(\boldsymbol{q}, t) \cdot \dot{\boldsymbol{q}} + \partial g / \partial t = 0$ which involves velocities. The definition of holonomic constraints thus includes equations which relate positions and momenta, but only where these are, through the differential equations, equivalent to equations involving positions only (4.3). For more details on constrained dynamics, see for example [67, 79, 83, 164, 227, 249].

An important type of holonomic constraint arising in molecular dynamics relates to bond vibration. Recall that the stretch of a chemical bond between two atoms gives rise to a potential energy contribution of the form

$$\varphi(r) = \frac{k}{2}(r - r_0)^2,$$

where r would typically represent the distance between a pair of atoms, say atoms i and j, $r = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|$. The strong chemical bond stretches involving hydrogens and various heavier atoms such as nitrogen and carbon give rise to the most rapid modes present in the atomic motion. As we have seen, a harmonic oscillator with frequency Ω, when solved using the Verlet method, has a stability restriction of the form

$$h < \frac{2}{\Omega}.$$

One approach to addressing these stiff terms is to use a multiple timestepping method. A more direct approach is to introduce constraints to simply remove the stiff bond stretches. This makes sense if (a) the motion of the constrained system can be simulated using larger timesteps and (b) the rigidification of certain vibrational terms does not significantly alter the thermodynamic or dynamic properties of interest to the modeller.

To illustrate the progression from vibrational model to constrained model, one may consider a single particle subject to a potential $U(\boldsymbol{q})$ as well as an additional stiff restraint. We suppose the restraint to have an associated coefficient $k = \varepsilon^{-1}$, where ε is a small parameter. Then we have a generalized flexible pendulum with Hamiltonian $H = \|\boldsymbol{p}\|^2/(2m) + U(\boldsymbol{q}) + \varepsilon^{-1}(\|q\| - r_0)^2/2$ (see the left panel of Fig. 4.4). The differential equations describing this model are

$$\begin{aligned} \dot{\boldsymbol{q}} &= m^{-1}\boldsymbol{p}, \\ \dot{\boldsymbol{p}} &= -\nabla U(\boldsymbol{q}) - \nabla \varphi_\varepsilon(\boldsymbol{q}), \end{aligned}$$

where the restraint potential $\varphi_\varepsilon(\boldsymbol{q}) = \varepsilon^{-1}(\|q\| - r_0)^2/2$ has associated restoring force

$$\boldsymbol{F}_\varepsilon = -\nabla \varphi_\varepsilon(\boldsymbol{q}) = \varphi_\varepsilon(\boldsymbol{q}).$$